Experiment – 4.1

Aim: Demonstration of COMMIT, SAVEPOINT and ROLLBACK in CLASS Table

ID	Name
1	Varun Chauhan
2	Jony Gautam
3	Dolly
4	Priya
5	Aman
6	Himanshi
7	Sakshi

```sql
CREATE DATABASE Varun;

USE Varun;

CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50)
);

INSERT INTO Student VALUES
(1, 'Varun'),
(2, 'Akansha'),
(3, 'Dolly'),
(4, 'Priya'),
(5, 'Aman'),
(6, 'Himanshi'),
(7, 'Sakshi');

COMMIT;
```

```sql
UPDATE Student SET Name='Abhijeet'
where Student_ID ='5';

SELECT * FROM STUDENT;

SAVEPOINT A;

INSERT INTO Student VALUES
(8,'Cheris');

SAVEPOINT B;

INSERT INTO Student VALUES
(9, 'Ram');

SAVEPOINT C;

SELECT * FROM STUDENT;

ROLLBACK TO B;

SELECT * FROM Student;

ROLLBACK TO A;
```

Output:

UPDATE class SET name = 'Abhijeet' WHERE id = '5';

ID 5 name changes from Rahul -> Abhijeet

1. SELECT 1 Output:

Student_ID	Name
1	Jony
2	Akansha
3	Dolly
4	Priya
5	Abhijeet
6	Himanshi
7	Sakshi
NULL	NULL

2. ROLLBACK TO B;

SELECT 2 Output:

Student_ID	Name
1	Jony
2	Akansha
3	Dolly
4	Priya
5	Abhijeet
6	Himanshi
7	Sakshi
NULL	NULL

Experiment – 4.2

Aim: To study and perform Transaction Control Language (TCL) commands such as SAVEPOINT, ROLLBACK, ROLLBACK TO SAVEPOINT, and COMMIT on the given table ACCOUNT_TXN.

Acc_ID	Holder_Name	Balance
101	Aman	5000
102	Bhavna	7000
103	Chirag	6000

Code:

```
CREATE DATABASE Account;

USE Account;

CREATE TABLE ACCOUNT_TXN (
    Acc_ID INT PRIMARY KEY,
    Holder_Name VARCHAR(50),
    Balance INT
);

INSERT INTO ACCOUNT_TXN VALUES
(101, 'Aman', 5000),
(102, 'Bhawna', 7000),
(103, 'Chirag', 6000);

INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

SAVEPOINT S1;

UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101;

UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102;

SAVEPOINT S2;

DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103;

INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);
```

```sql
SAVEPOINT S3;

UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S2;

SELECT * FROM ACCOUNT_TXN;

INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500);

SAVEPOINT S4;

UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN;

ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

COMMIT;
```

Output:

1. SELECT 1 Output:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna Sharma	6000
104	Diya	4000
105	Eshan	3000
NULL	NULL	NULL

2. Select 2 Output (After ROLLBACK TO S2):

Result Grid | Filter Rows:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhawna	6000
103	Chirag	6000
104	Diya	4000
NULL	NULL	NULL

3. Select 3 output:

Result Grid | Filter Rows:

Acc_ID	Holder_Name	Balance
101	Aman	7000
102	Bhawna	6000
103	Chirag	6000
104	Diya	4500
106	Farah	4500
NULL	NULL	NULL

4. Select 4 Output (After ROLLBACK TO S4):

Result Grid | Filter Rows:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhawna	6000
103	Chirag	6000
104	Diya	4000
106	Farah	4500
NULL	NULL	NULL

5. Rows Permanently Saved After COMMIT:

Acc_ID	Holder_Name	Balance
101	Aman	6500
102	Bhavna	6000
103	Chirag	6000
104	Diya	4000
105	Farah	4500

1. Difference between ROLLBACK and ROLLBACK TO SAVEPOINT

- ROLLBACK cancels the entire transaction.
- ROLLBACK TO SAVEPOINT cancels changes only up to a specific savepoint.

2. Can we rollback to S3 after ROLLBACK TO S2?

- No.
  Because when we rollback to S2, all savepoints created after S2 (like S3) are removed.

3. What happens if COMMIT is executed before creating any savepoint?

- All changes become permanent and cannot be rolled back.